Jae Ho Choi

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EMPLOYMENT

Department of Mathematical Sciences, Carnegie Mellon University, Pittsburgh, Pennsylvania, USA

- Postdoctoral Research Associate — Sep 2024 – Present

EDUCATION

University of Pennsylvania, Philadelphia, Pennsylvania, USA

- Ph.D. in Mathematics — Sep 2019 – May 2024
  - Advisors: Robert Strain, Yoichiro Mori
- Master of Computer and Information Technology (MCIT) — Sep 2021 – May 2023

University of Maine, Orono, Maine, USA

- M.A. in Mathematics — Sep 2017 – May 2019
  - Thesis: Theory of Lexicographic Differentiation in the Banach Space Setting
  - Adsivor: Peter Stechlinski

Williams College, Williamstown, Massachusetts, USA

- B.A. in Physics, Mathematics, and Economics — Sep 2013 – May 2017
  - Senior Thesis (in Physics): An Unusual Derivation of the Boltzmann Distribution for a Particle in a 1D Box
  - Advisor: William K. Wootters
  - Graduated with Departmental Honors in Physics

CITIZENSHIP

South Korean, with U.S. permanent residency

PERSONAL WEBSITE

choijh93.github.io

RESEARCH INTERESTS

partial differential equations arising in fluid dynamics, primarily analytical study of stability, long-time dynamics, and well-posedness of free boundary problems; numerical simulation of free boundary problems

PREPRINTS

[9] J. Choi and I. Tice, Traveling wave solutions to a two-phase Stokes problem with viscosity contrast driven by surface tension in 2D, *in preparation*.

[8] J. Choi and R. Strain, Stability of a two-phase Taylor-Melcher problem with viscosity contrast driven by surface tension in 2D, *in preparation*.

[7] J. Choi, Stability of a two-phase Brinkman problem with surface tension in 2D, *in preparation*.

[6] L. Bouck, J. Choi, and N. Walkington, A stable boundary integral scheme for a two-phase Brinkman problem in 2D, *in preparation*.

[5] J. Choi and R. Strain, Stability of a two-phase Stokes problem with viscosity contrast driven by surface tension in 2D, *in preparation*.

[4] J. Choi and I. Tice, Traveling wave solutions to a general incompressible Navier-Stokes-Fourier system with free boundary, *submitted*.

[3] J. Choi, Stability of a two-phase Stokes problem with surface tension, *submitted*.

PUBLICATIONS

[2] J. Choi, N. Krishna, N. Magill, and A. Sarria, On the L^p regularity of solutions to the generalized Hunter-Saxton system, *Discrete Contin. Dyn. Syst. Ser. B* 24 (2019), 6349-6365.

[1] S. Alterman, J. Choi, R. Durst, S. Fleming, and W. Wootters, The Boltzmann distribution and the quantum-classical correspondence, *J. Phys. A* 51 (2018), no. 34.

INVITED TALKS

- *A two-phase Stokes problem with viscosity contrast*.
  Graduate Student and Postdoc (GSPS) Seminar
  Department of Mathematical Sciences, Carnegie Mellon University, PA, USA — Oct 2025
- *On the two-phase Stokes flow problem with surface tension*.
  Recent Progress in PDE Models of Incompressible Fluids
  AMS Western Sectional Meeting, University of Denver, CO, USA — Aug 2025

- *Stability and instantaneous analyticity of a 2D Stokes bubble with viscosity contrast*.
  PDE and Applied Analysis Seminar
  Pohang University of Science and Technology, Pohang, Republic of Korea — Jul 2025
- *Well-posedness of a two-phase Stokes flow problem with surface tension*.
  Center for Nonlinear Analysis (CNA) Seminar
  Carnegie Mellon University, PA, USA — Sep 2024
- *On the two-phase Stokes flow problem with surface tension*.
  PDE and Applied Analysis Seminar
  Pohang University of Science and Technology, Pohang, Republic of Korea — Jun 2024
- *The mathematics of fluids*.
  UP GRADe (an NSF-funded talk series co-hosted by Penn's AWM chapter and GeMs in Math)
  Department of Mathematics, University of Pennsylvania, PA, USA — May 2024
- *The boundary integral method for interfacial Stokes flow*.
  Analysis Seminar, Department of Mathematics, University of Pennsylvania, PA, USA — Apr 2024
- *On the two-phase Stokes flow problem with surface tension*.
  Analysis Seminar, Department of Mathematics, University of Pennsylvania, PA, USA — Nov 2023
- *On the two-phase Stokes flow problem with surface tension*.
  Applied and Numerical PDE Seminar, University of California, Berkeley, CA, USA — Nov 2023
- *A friendly introduction to the Taylor-Melcher model*.
  Recent Advances in Mathematical Fluid Dynamics, Duke University, NC, USA — May 2023
- *Infinite-energy solutions of the Euler equations with time-dependent damping*.
  Williams College REU Conference, Williams College, MA, USA — Jul 2016
- *Reduction of error in the discrete truncated Wigner approximation*.
  Williams College Summer Science Research Poster Session, Williams College, MA, USA — Aug 2015

INVITED CONFERENCES

- POSTECH-KIAS Intensive Lecture Series on PDEs I & II, Republic of Korea — Jun 2026
- Mathematics of General Relativity and Fluids (SLMath), Heraklion, Crete, Greece — Jul 2024
- Summer School on Wave Turbulence, Massachusetts Institute of Technology, MA, USA — Jul 2023
- New Trends in Mathematical Fluid Dynamics, Institut Fourier, Gières, France — Jun 2023
- Recent Advances in Mathematical Fluid Dynamics, Duke University, NC, USA — May 2023

SERVICE & STUDENT MENTORSHIP

- Panelist, Graduate School Panel (hosted by the Student Mathematics and Statistics Advisory Board)
  Department of Mathematics, Williams College, MA, USA — Oct 2024
- Mentor, Directed Reading Program (DRP), Department of Mathematics, University of Pennsylvania
  DRP is a program in which undergraduate students are paired with graduate student mentors for independent study based on matching interests.
  - Darren Zheng, harmonic analysis (Spring 2024).
  - Dylan Marchlinski, fluid dynamics and probability theory (Fall 2023).
  - Evan Qiang, probability theory (Spring 2023).
  - David Kogan, probability theory (Spring 2022).

TEACHING EXPERIENCE

CARNEGIE MELLON UNIVERSITY

- Instructor, Integration and Approximation (21-122) — Fall 2024
- Instructor, Integration and Approximation (21-122) — Spring 2025
- Instructor, Integration and Approximation (21-122) — Fall 2025
- Co-Instructor, Integration and Approximation (21-122) — Spring 2026

UNIVERSITY OF PENNSYLVANIA

- Teaching Assistant, Advanced Analysis (MATH 5090) — Spring 2024
- Teaching Assistant, Advanced Analysis (MATH 5080) — Fall 2023
- Grader, Mathematics in the Age of Information (MATH 2100) — Spring 2022
- Teaching Assistant, Ideas in Mathematics (MATH 1700) — Spring 2021
- Teaching Assistant, Advanced Calculus (MATH 3610) — Fall 2020

UNIVERSITY OF MAINE

- Teaching Assistant, Calculus II (MAT 127) — Spring 2019
- Teaching Assistant, Calculus I (MAT 126) — Fall 2018
- Teaching Assistant, Calculus II (MAT 127) — Spring 2018

- Teaching Assistant, Calculus II (MAT 127) — Fall 2017

WILLIAMS COLLEGE

- Teaching Assistant, Real Analysis (MATH 350) — Spring 2016
- Teaching Assistant, Linear Algebra (MATH 250) — Fall 2015

AWARDS & SCHOLARSHIPS

- **Good Teaching Award**
  Department of Mathematics, University of Pennsylvania — Fall 2023, Spring 2024
- **Benjamin Franklin Fellowship**
  Department of Mathematics, University of Pennsylvania — Sep 2019 – May 2024
- **Class of 1960 Scholar in Physics**
  Department of Physics, Williams College — Sep 2016 – May 2017
- **John & Louise Finnerty Class of 1971 Fund for Applied Mathematical Research**
  Department of Mathematics and Statistics, Williams College — Jun 2016– Aug 2016